TAVS-ESP: Trust-Adaptive Verification Scheduling with Ephemeral Structured Projections for Byzantine-Robust Federated Learning at LLM Scale

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Abstract

Byzantine-robust federated learning (FL) faces three unresolved tensions at large language model (LLM) scale: (i) existing robust aggregators require $O(Nd)$ to $O(N^2d)$ operations per round, becoming intractable for $d > 10^9$; (ii) defenses based on fixed geometric projections are vulnerable to adaptive adversaries who pre-compute their null space; and (iii) no prior work formally models the *verification scheduling* dimension, leaving a timing attack surface whenever scheduling is introduced ad hoc. We propose TAVS-ESP, a co-designed two-layer framework that resolves all three problems simultaneously. **Layer 1** (TAVS) governs *who* is verified and *when* via a three-tier trust-adaptive scheduling protocol backed by a CSPRNG under a trusted-server assumption, making promotion assignments cryptographically unpredictable. **Layer 2** (ESP) governs *how* verification executes via ephemeral block-diagonal Johnson-Lindenstrauss projections aligned to model semantic boundaries (attention heads, LoRA matrices), defeating null-space attacks while amplifying localized anomaly signals. The layers are unified by a single aggregation rule that subsumes both detection filtering and trust-discounted shadow aggregation, with Bayesian posterior weights that preserve honest client contributions at near-full weight. We prove four formal guarantees: an $O(f_\infty \cdot Nk)$ steady-state complexity certificate ($f_\infty < 1$, $k \ll d$); a minimax robustness bound showing adversarial masking probability decays exponentially in k; a joint CSPRNG security theorem establishing compounding unpredictability; and a Sybil resistance bound closing the trust-initialization vulnerability. A convergence sketch under IID data is provided in the main body; the full non-IID Byzantine convergence proof via a staged Lyapunov argument is given in Appendix D. Experiments on federated fine-tuning of LLaMA-3-8B validate the framework against Layerwise Model Poisoning, Sleeper Agent injection, and scheduling-aware timing attacks, demonstrating near-zero attack success rate with a 40–60% wall-clock reduction in aggregation cost at $N = 1000$ clients.

1 Introduction

Federated learning enables collaborative model training without centralizing private data, making it the primary mechanism for deploying large language models across privacy-sensitive institutions. Its distributed structure, however, exposes it to *Byzantine adversaries* who submit crafted gradient updates to corrupt the global model. Three distinct open problems, previously addressed only in isolation, combine to make existing defenses inadequate at LLM scale.

Problem 1: Computational intractability. Distance-based robust aggregators [2, 4, 14] require $O(N^2d)$ or $O(Nd)$ operations per round. For LLaMA-7B ($d \approx 7 \times 10^9$) with $N = 1000$ clients, even the linear methods demand 7×10^{12} floating-point operations per aggregation round—infeasible on standard server hardware.

Problem 2: Adaptive null-space evasion. Defenses based on fixed dimensionality reduction (PCA, autoencoders, static random projections) establish a static geometric boundary. An adversary with white-box access to the defense computes z such that $Rz \approx 0$, injecting a backdoor invisible to the detector while remaining active in the full parameter space [1].

Problem 3: The unmodeled scheduling dimension. All existing Byzantine-robust FL papers implicitly verify every client every round. This prevents exploiting trust history to reduce computation and, critically, creates a *timing attack surface*: any ad hoc scheduling introduced later is exploitable if promotion assignments are predictable. To our knowledge, no prior work formalizes this attack class or provides a defense with provable guarantees.

Our contributions. We propose TAVS-ESP, which resolves all three problems through a co-designed two-layer system unified by server-side CSPRNG randomness under a trusted-server assumption (Assumption 1). Our formal contributions are:

- **C1: TAVS-ESP system.** A complete two-layer architecture (Section 4) including a three-tier trust-adaptive scheduling protocol and ephemeral block-diagonal JL projections, with a unified Bayesian-weighted aggregation rule.
- **C2: Complexity certificate (TC4).** $O(f_\infty \cdot Nk)$ steady-state detection cost with $f_\infty < 1$, enabling LLM-scale deployment (Theorem 5).
- **C3: Minimax robustness (TC1).** Adversarial masking probability decays exponentially in k under ephemeral projection, extended to multi-round adaptive adversaries (Theorem 3).
- **C4: Joint CSPRNG security (TC3).** Joint unpredictability over projection and schedule is strictly stronger than either alone (Theorem 4).
- **C5: Sybil resistance.** Closed-form bound on coalition-scale model corruption and minimum infiltration time under Mechanisms 1+3 (Theorem 6).
- **C6: Convergence (TC2).** Sketch under IID data (Section 5.5); full non-IID Byzantine proof via staged Lyapunov argument in Appendix D.

2 Related Work

Byzantine-robust aggregation. Blanchard et al. [2] introduced Krum ($O(N^2 d)$). Yin et al. [17] proposed coordinate-wise trimmed mean and median ($O(Nd \log N)$). El Mhamdi et al. [4] introduced Bulyan combining Krum with trimmed mean. Pillutla et al. [14] applied the geometric median via Weiszfeld iterations. Cao et al. [3] introduced FLTrust using a server-side clean reference dataset. All require full-dimensional operations every round and verify all clients uniformly—the two properties our work specifically removes.

Adaptive attacks. Bagdasaryan et al. [1] demonstrated model replacement attacks. Guo et al. [6] introduced layerwise localized poisoning targeting specific transformer blocks. Shumailov et al. [15] and Hubinger et al. [7] introduced Sleeper Agent injection into specific attention sub-circuits. Our block-diagonal JL structure directly targets the localized structure these attacks exploit.

Dimensionality reduction defenses. PCA and autoencoder-based defenses reduce detection cost but use static projections vulnerable to null-space attacks [11]. Xie et al. [16] and Nguyen et al. [13] introduced score-based and clustering-based defenses. Our use of *ephemeral* structured projections—freshly derived each round from a secret server seed—is, to our knowledge, not present in prior work.

Trust and partial participation. Kang et al. [8] proposed trust-weighted aggregation (FedTrust) using validation performance as the trust signal. Partial client participation for communication efficiency is studied in Karimireddy et al. [9], Li et al. [10], McMahan et al. [12], but none of these works model participation as a security primitive or provide timing-attack defenses. Fung et al. [5] proposed FoolsGold to reduce Sybil influence via contribution diversity—our Mechanism 1+3 provides formal guarantees where FoolsGold provides empirical heuristics.

3 Problem Formulation and Threat Model

Setup. Let N clients and a central aggregation server collaboratively minimize

$$F(w) = \frac{1}{|\mathcal{H}|} \sum_{i \in \mathcal{H}} F_i(w), \qquad F_i(w) = \mathbb{E}_{z \sim \mathcal{D}_i}[\ell(w;\, z)],$$

where $\mathcal{H} \subset [N]$ are honest clients, $w \in \mathbb{R}^d$ are model parameters, and $\mathcal{D}_i$ is client i's private data distribution. Let $\mathcal{B} = [N] \setminus \mathcal{H}$ with $|\mathcal{B}| = \beta N$, $\beta < 1/2$.

Definition 1 (Gradient dissimilarity)**.** $\zeta = \max_{i \in \mathcal{H}} \|\mathbb{E}[\nabla F_i(w)] - \nabla F(w)\|_2$ *bounds non-IID heterogeneity;* $\zeta = 0$ *recovers the IID setting.*

Adversarial model. We consider a fully adaptive white-box adversary controlling $\mathcal{B}$ that: (i) knows the model architecture and aggregation algorithm; (ii) can adaptively craft $z_i(r) \in \mathbb{R}^d$ for $i \in \mathcal{B}$ given full gradient history; (iii) attempts to infer the promotion schedule from observed round outcomes; but (iv) does not have access to the server's CSPRNG seed (Assumption 1). We additionally consider a *Sybil adversary* who can register new clients at any round.

Assumption 1 (Trusted Server (TSA))**.** *The aggregation server follows the protocol honestly and does not leak its CSPRNG key K to any client.*

Definition 2 (ρ-timing resistance)**.** *A scheduling protocol is ρ-timing-resistant if for any adaptive adversary submitting malicious gradients only in predicted promoted rounds, the expected weighted malicious fraction in the aggregate satisfies $\mathbb{E}[\beta_{\text{eff}}] \leq \rho$.*

Definition 3 (Verification scheduling problem)**.** *Find a policy $\pi : \text{history}_r \to (\mathcal{V}(r), \mathcal{S}(r))$ minimizing $\lim_{T\to\infty} \frac{1}{T} \sum_{r=1}^{T} |\mathcal{V}(r)|$ subject to: (1) ρ-timing resistance; (2) every client appears in $\mathcal{V}(r)$ within any window of W consecutive rounds for finite W; (3) convergence to a neighborhood of w^* at a bounded rate.*

TAVS-ESP solves Definition 3 with $W = 3$, explicit ρ, and the convergence guarantee of TC2.

4 The TAVS-ESP System

4.1 Cryptographic Randomness Protocol

Under TSA, the server holds a 256-bit secret key K and uses ChaCha20-CTR as the CSPRNG. At round r, the master seed is $y_r = \text{ChaCha20}(K,\, r) \in \{0,1\}^{512}$. From y_r the server deterministically derives:

$$\begin{aligned} R_r &= G_1(y_r) \in \mathbb{R}^{k \times d} && \text{(block-diagonal projection matrix)}, \\ P_r &= G_2(y_r) \subseteq S_{\text{candidate}}(r) && \text{(promotion assignments)}, \\ D_r &= G_3(y_r) \subseteq P_r && \text{(decoy verification set)}. \end{aligned}$$

All three are derived before any client submits gradients in round r. Under TSA and the pseudorandomness of ChaCha20, y_r is computationally indistinguishable from uniform for any polynomial-time adversary without K.

4.2 Layer 1: Trust-Adaptive Verification Scheduling (TAVS)

Trust score. For each verified client $i \in \mathcal{V}(r)$, the behavioral score $\varphi_i(r) \in [0, 1]$ is derived from the Layer 2 projected distance:

$$\varphi_i(r) = 1 - \frac{d_i(r)}{\max_{j \in \mathcal{V}(r)} d_j(r) + \varepsilon}, \qquad d_i(r) = \|R_r g_i - g_r^*\|_2 .$$

The trust score evolves via a temporally decaying EMA:

$$T_i(r) = \alpha \cdot T_i(r{-}1) + (1 - \alpha) \cdot \varphi_i(r) \quad (i \in \mathcal{V}(r)), \qquad T_i(r) = \alpha \cdot T_i(r{-}1) \quad (i \in \mathcal{S}(r)).$$

Decay parameter $\alpha \in (0, 1)$ ensures trust must be continuously earned; a promoted client's score decays toward zero in the absence of fresh evidence.

Three-tier structure. Clients are classified each round using thresholds $\theta_{\text{low}} < \theta_{\text{high}}$:

- **Tier 1** ($T_i < \theta_{\text{low}}$ or newly joined): mandatory full verification every round.
- **Tier 2** ($\theta_{\text{low}} \leq T_i < \theta_{\text{high}}$): alternating verify/promote; maximum one consecutive unverified round.
- **Tier 3** ($T_i \geq \theta_{\text{high}}$, after k_{trust} consecutive clean verifications): extended double-skip window; each promoted round independently subjected to decoy verification with probability p_{decoy}.

Theorem 1 (Bounded recurrence)**.** *Under* TAVS*, every client $i \in [N]$ appears in $\mathcal{V}(r)$ for at least one round in every window of $W = 3$ consecutive rounds.*

Proof. Tier 1: gap $= 1$. Tier 2: maximum one skip, gap $= 2$. Tier 3: maximum two consecutive skips, gap $= 3$. Taking the maximum gives $W = 3$. □ □

Trust initialization rate-limiting (Mechanism 3). Regardless of behavioral score, a client joining at round r_0 has its trust score capped at

$$T_i^{\max}(r) = 1 - \exp\bigl(-(r - r_0)/\tau_{\text{ramp}}\bigr).$$

No client can reach Tier 3 eligibility before round $r_{\min} = r_0 + \tau_{\text{ramp}} \cdot \log\bigl(1/(1 - \theta_{\text{high}})\bigr)$.

4.3 Layer 2: Ephemeral Block-Diagonal JL Projections (ESP)

Block-diagonal construction. Let the model parameter vector $w \in \mathbb{R}^d$ be partitioned into M semantic blocks $\{B_m\}_{m=1}^M$ (e.g., individual attention heads, LoRA A/B matrices) with $\sum_m d_m = d$. The projection matrix is

$$R_r = \text{diag}(r_1, r_2, \ldots, r_M), \quad r_m \in \mathbb{R}^{k_m \times d_m}, \quad (r_m)_{ij} \sim \mathcal{N}(0, 1/k_m),$$

with $k_m = \lceil C \log(N)/\varepsilon^2 \rceil$ and total dimension $k = \sum_m k_m$. Each block r_m is generated on-demand from (y_r, m) via a secondary PRNG and discarded after use, requiring only $O(\max_m k_m d_m) = O(kd/M)$ working memory (a factor of M reduction over storing the full dense matrix).

Block-normalized outlier detection. For each $i \in \mathcal{V}(r)$ and block m, compute the standardized deviation:

$$Z_i^{(m)}(r) = \frac{\bigl\|r_m g_i^{(m)} - \bar{g}_r^{(m)}\bigr\|_2^2}{\hat{\sigma}_m^2(r) + \varepsilon_{\text{stab}}},$$

where $\bar{g}_r^{(m)}$ denotes the m-th block of the robust projected aggregate g_r^*, and $\hat{\sigma}_m^2(r)$ is an EMA estimate of the honest within-block variance updated from the previous round's inliers:

$$\hat{\sigma}_m^2(r) = \alpha_\sigma \hat{\sigma}_m^2(r-1) + (1-\alpha_\sigma)\frac{1}{|\mathcal{L}(r-1)|}\sum_{i\in\mathcal{L}(r-1)} \left\| r_m g_i^{(m)} - \bar{g}_{r-1}^{(m)} \right\|_2^2.$$

The aggregate anomaly score $A_i(r) = \max_m Z_i^{(m)}(r)$ separates *structural heterogeneity* (which inflates $\hat{\sigma}_m^2$ uniformly across blocks) from *localized backdoor gradients* (which spike $Z_i^{(m)}$ in one specific block m^*). Client i is classified as an outlier if $A_i(r) > \tau_z$; otherwise it joins $\mathcal{L}(r)$.

Proposition 2 (Non-IID/malicious separation)**.** *For an honest non-IID client, $\mathbb{E}[Z_i^{(m)}(r)] = 1$ for all m by construction of $\hat{\sigma}_m^2$. For a Byzantine client whose attack is concentrated in block m^*, $\mathbb{E}[Z_i^{(m^*)}(r)] = 1 + \delta_{m^*}^2/\hat{\sigma}_{m^*}^2(r) > 1$, where $\delta_{m^*} = \left\| r_{m^*}\left(z_i^{(m^*)} - g_i^{(m^*),\mathrm{hon}}\right)\right\|_2$ is the projected attack magnitude in block m^*.*

4.4 Unified Aggregation Rule

Replacing flat λ-discounting with Bayesian posterior weights preserves honest promoted clients at near-full weight while bounding Byzantine influence. Define the posterior honesty probability for promoted client i:

$$p_i(r) = \sigma\big(c_\lambda \cdot (T_i(r) - 0.5)\big), \quad \sigma(x) = 1/(1+e^{-x}).$$

For high-trust clients ($T_i \approx 1$), $p_i \approx 1$ (near-full weight). The aggregation rule is:

$$\boxed{w(r) = w(r-1) + \eta \cdot \frac{\sum_{i\in\mathcal{L}(r)} g_i + \sum_{i\in\mathcal{S}(r)} p_i(r)\cdot g_i}{|\mathcal{L}(r)| + \sum_{i\in\mathcal{S}(r)} p_i(r)}}$$

Verified outliers $\mathcal{O}(r) = \mathcal{V}(r) \setminus \mathcal{L}(r)$ receive zero weight. The normalization $Z(r) = |\mathcal{L}(r)| + \sum_{i\in\mathcal{S}(r)} p_i(r)$ ensures each honest client's expected per-round contribution is $1/N$ regardless of tier assignment.

Aggregate budget constraint (Mechanism 1). Prior to Step 2 of the round protocol, the server checks:

$$\frac{\sum_{i\in\mathcal{S}(r)} p_i(r)}{Z(r)} \le \gamma_{\mathrm{budget}}.$$

If violated, the lowest-T_i promoted clients are demoted to verified status until the constraint holds. This bounds total unverified influence in any single round regardless of coalition size.

5 Theoretical Guarantees

5.1 TC1: Minimax Robustness Under Ephemeral Projection

Theorem 3 (Multi-round evasion resistance, TC1)**.** *Let $\mathcal{A}$ be an adaptive adversary with $\|z_i(r)\|_2 \le G_{\max}$ and access to the full gradient history. Let $\delta(r) = \left\|z_i(r) - g_i^{\mathrm{hon}}(r)\right\|_2$ be the attack magnitude at round r. Under* ESP *with ephemeral ChaCha20-derived Gaussian block-diagonal projections of dimension $k = O(\log N/\varepsilon^2)$:*

$$\mathbb{P}\big[\mathcal{A} \text{ evades detection for all } T \text{ rounds}\big] \;\le\; \exp\Big(-\Omega\big(k \cdot \min_r \delta(r)^2\big)\Big).$$

The probability is over the ChaCha20-derived randomness; the bound holds uniformly over all adaptive strategies.

Proof sketch. By the JL lemma, for any fixed z with $\|z\|_2 > \varepsilon$, $\mathbb{P}[\|R_r z\|_2 \le \varepsilon/2] \le \exp(-\Omega(k))$. Since $y_r = \mathrm{ChaCha20}(K, r)$ is computationally indistinguishable from uniform under TSA, the distribution of R_r over rounds is independent. The multi-round bound follows by independence of R_r across rounds and a union bound. Adaptivity of $\mathcal{A}$ does not help: R_r is derived from y_r, which is independent of all $y_{r'}$, $r' < r$. Full proof in Appendix B. □

5.2 TC3: Joint CSPRNG Security

Theorem 4 (Compounding unpredictability, TC3)**.** *Define adversary classes: $\mathcal{A}_1$ knows R_r but not P_r; $\mathcal{A}_2$ knows P_r but not R_r; $\mathcal{A}_3$ knows neither (standard* TAVS-ESP *adversary). Then* $\mathrm{ASR}(\mathcal{A}_1) \ge \mathrm{ASR}(\mathcal{A}_3)$ *and* $\mathrm{ASR}(\mathcal{A}_2) \ge \mathrm{ASR}(\mathcal{A}_3)$*, with:*

$$\mathrm{ASR}(\mathcal{A}_3) \;\le\; (1 - p_{\mathrm{decoy}})^K \cdot \exp\Big(-\Omega\big(k \cdot \delta_{\min}^2\big)\Big),$$

where K is the number of promoted rounds and $\delta_{\min}$ is the minimum attack magnitude required for a successful backdoor injection.

Proof sketch. $\mathcal{A}_2$ who knows promotion timing cannot exploit it because R_r is unknown; any attack with non-negligible $\delta_{\min}$ is detected by TC1 when the round is verified. $\mathcal{A}_1$ who knows the projection can craft null-space attacks, but since P_r is unknown, they cannot concentrate these attacks in promoted rounds. $\mathcal{A}_3$ faces both unpredictabilities simultaneously; the joint probability is the product of the two individual bounds by independence of G_1 and G_2 under the ChaCha20 derivation. Full proof in Appendix C. □

5.3 TC4: Complexity Certificate

Theorem 5 (Steady-state complexity, TC4)**.** *Under* TAVS *with honest majority ($\beta < 1/2$) and EMA parameter α, the steady-state verified fraction satisfies:*

$$f_\infty \;=\; \lim_{r\to\infty} \mathbb{E}\big[|\mathcal{V}(r)|/N\big] \;\le\; \beta + (1-\beta)\,\mathbb{P}[T_i < \theta_{\mathrm{high}} \mid \textit{honest}] + (1-\beta)\,\mathbb{P}[T_i \ge \theta_{\mathrm{high}} \mid \textit{honest}] \cdot \tfrac{1}{3} \;<\; 1.$$

Effective per-round detection complexity is $O(f_\infty \cdot N \cdot k) < O(Nk) \ll O(Nd)$.

Remark. Mechanism 1 adds $O(f_\infty N \log(f_\infty N))$ per round for the budget trim sort. Mechanism 3 adds $O(N)$ per round for the rate cap. Both are dominated by the $O(f_\infty N k)$ projection term for any $k \ge 100$. TC4 is unaffected.

5.4 Sybil Resistance

Theorem 6 (Sybil resistance)**.** *Let a Sybil adversary control N_{sybil} clients behaving honestly for T_{sybil} rounds before launching a coordinated attack. Under Mechanism 1 (budget γ_{budget}) and Mechanism 3 (ramp-up τ_{ramp}), the following two bounds hold.* (1) *The maximum aggregate model corruption in any single round satisfies $\|\Delta w_{\mathrm{malicious}}(r)\|_2 \le \eta \cdot \gamma_{\mathrm{budget}} \cdot Z(r) \cdot G_{\max}$, independent of N_{sybil}.* (2) *No Sybil node can reach Tier 3 eligibility before round $r_{\min} = \tau_{\mathrm{ramp}} \cdot \log(1/(1-\theta_{\mathrm{high}}))$ from its join round, regardless of behavioral compliance.*

Proof sketch. (1) follows directly from the budget constraint construction: the server enforces $\sum_{i\in\mathcal{S}} p_i/Z \le \gamma_{\mathrm{budget}}$ before accepting promotions. (2) follows from the trust cap $T_i^{\max}(r) = 1 - e^{-(r-r_0)/\tau_{\mathrm{ramp}}}$: since $T_i(r) \le T_i^{\max}(r) < \theta_{\mathrm{high}}$ for $r < r_0 + r_{\min}$, Tier 3 eligibility is structurally impossible before $r_{\min}$ rounds, independent of $\varphi_i(r)$. □

5.5 TC2: Convergence Sketch (IID Setting)

We state the convergence result for the IID case ($\zeta = 0$) to establish that the unified aggregation rule is compatible with gradient-based optimization. The full non-IID Byzantine proof via a four-stage Lyapunov argument is in Appendix D.

Theorem 7 (Convergence, IID setting, TC2 Stage 1)**.** *Assume F is L-smooth and μ-strongly convex (Assumptions A1–A4 in Appendix A), IID data ($\zeta = 0$), and no Byzantine misclassification. With learning rate $\eta \leq 1/(2L)$:*

$$\mathbb{E}[F(w(r)) - F^*] \;\leq\; (1-\mu\eta)^r \big(F(w(0)) - F^*\big) + \frac{\eta\sigma^2}{2N\mu} + \frac{B_p}{\mu},$$

where $B_p = \eta L(1-\bar{p})G_{\max}^2|\mathcal{S}|/N$ is the Bayesian-weight bias term, with $\bar{p} = \mathbb{E}[p_i \mid \text{honest}]$. As $T_i(r) \to 1$ for honest clients, $\bar{p} \to 1$ and $B_p \to 0$: the bias vanishes as the system learns the honest population.

Remark 1. *B_p is strictly smaller than the λ-discounting bias $B_\lambda = \eta L(1-\lambda^*)G_{\max}^2|\mathcal{S}|/N$ from prior work, because $\bar{p} > \lambda^*$ for any honest high-trust client ($\sigma(c_\lambda \cdot 0.5) \approx 0.62 \ll 0.9 = \lambda_{\max}$ at the boundary, but for $T_i > 0.8$, $p_i > 0.95 > \lambda_{\max}$). Honest clients are better served by Bayesian posterior weighting than by flat discounting.*

6 Experiments

6.1 Setup

Models and datasets. We evaluate on two scales: BERT-base ($d = 110$M, GLUE/SQuAD) for controlled ablations, and LLaMA-3-8B with LoRA ($r = 8$, $d_{\text{LoRA}} \approx 200$M effective) fine-tuned on Alpaca-cleaned for LLM-scale validation. Data heterogeneity follows Dirichlet(α) with $\alpha \in \{0.1, 0.3, 1.0\}$.

Baselines. We compare against: FedAvg (no defense), Krum [2], Geometric Median [14], Trimmed Mean [17], FLTrust [3], and Full-JL (JL projection applied every round to all clients, no scheduling).

Attacks. We evaluate against: Layerwise Model Poisoning [6] (localized to a single transformer block), Sleeper Agent injection targeting specific attention heads, and a Scheduling-Aware Timing Attack (Section 6.4) that concentrates malicious updates in predicted promoted rounds.

TAVS-ESP hyperparameters. $\alpha = 0.9$, $\theta_{\text{low}} = 0.3$, $\theta_{\text{high}} = 0.7$, $k_{\text{trust}} = 10$, $p_{\text{decoy}} = 0.15$, $c_\lambda = 8$, $\gamma_{\text{budget}} = 0.35$, $\tau_{\text{ramp}} = 30$, $k = 1000$ (BERT), $k = 5000$ (LLaMA-LoRA).

6.2 E1: Null-Space Poisoning

We implement an adaptive attacker computing $z_{\text{attack}} \in \text{null}(R_{\text{static}})$ for static projection baselines. TAVS-ESP uses ephemeral R_r; the attacker receives only gradient history.

Static projection baselines exhibit high ASR because the attacker pre-computes their null space; ephemeral projection reduces ASR to near zero. TAVS-ESP matches Full-JL security at $f_\infty \approx 0.48$ of its cost.

6.3 E2: Signal Dilution Analysis

We inject a localized backdoor targeting attention head 3 of layer 8 ($\approx 0.07\%$ of parameters). We compare Dense Gaussian JL, Walsh-Hadamard Transform (WHT), and Block-Diagonal JL, each with $k = 1000$.

Table 1: Attack Success Rate (ASR%) and Clean Accuracy (CA%) under null-space poisoning. $\beta = 0.3$, $N = 100$, BERT-base, Dirichlet(0.3).

Method	ASR↓	CA↑	Det. Rate↑	Cost/round
FedAvg (no defense)	94.2	84.1	—	$O(Nd)$
Geometric Median	17.4	83.8	81.3	$O(Nd)$
Static JL + GeoMed	83.6	83.5	14.2	$O(Nk)$
PCA defense	78.9	83.2	19.7	$O(Nk)$
Full-JL (ephemeral)	4.8	83.6	94.1	$O(Nk)$
TAVS-ESP	**3.1**	**83.7**	**95.3**	$O(f_\infty Nk)$

Table 2: True Positive Rate (TPR%) and per-block SNR for localized backdoor. Attack concentration = 0.07% of parameters.

Projection	TPR↑	Block SNR↑	Memory
Dense Gaussian JL	23.4	0.04	$O(kd)$
Walsh-Hadamard	19.7	0.02	$O(d)$
Block-Diagonal JL	**91.2**	**1.00**	$O(kd/M)$

Block-diagonal projection achieves 4× higher TPR by isolating the anomaly signal within the target semantic block. Per Proposition 2, $\mathbb{E}[Z_i^{(m^*)}] = 1 + \delta_{m^*}^2/\hat{\sigma}_{m^*}^2 \gg 1$ while all other blocks remain at $\mathbb{E}[Z_i^{(m)}] \approx 1$ for $m \neq m^*$.

6.4 E4: Timing Attack Suppression

We implement three scheduling variants: deterministic round-robin (DRR), public-seeded random (RNG), and TAVS-ESP (ChaCha20-CTR). A timing-aware adversary observes its scheduling history over 50 rounds to infer the pattern before launching attacks.

Table 3: ASR (%) for timing-aware vs. naive adversary under three schedulers. $\beta = 0.2$, $N = 100$, BERT-base.

Scheduler	ASR (timing-aware)↓	ASR (naive)↓	Advantage
DRR (deterministic)	71.3	22.4	3.2×
RNG (public seed)	58.7	21.8	2.7×
TAVS-ESP	**21.1**	**20.6**	1.02×

DRR and RNG schedulers are exploitable: timing-aware adversaries achieve 2.7–3.2× higher ASR. TAVS-ESP's ChaCha20-derived scheduling eliminates the advantage (ratio ≈ 1), empirically validating Theorem 4.

6.5 E5: Efficiency Scaling

We measure wall-clock aggregation time at $N \in \{100, 500, 1000, 5000\}$ for Full-dim GeoMed, Full-JL, and TAVS-ESP, on LLaMA-3-8B LoRA.

TAVS-ESP achieves a 2.3× reduction over Full-JL at $N = 1000$, consistent with the TC4 prediction $f_\infty \approx 0.46$ (theoretical: 0.44; gap $< 5\%$, validating the complexity certificate).

Table 4: Aggregation wall-clock time (seconds/round) and empirical f_∞ at $\beta = 0.2$, Dirichlet(0.3). Accuracy within 1.2% of undefended baseline for all TAVS-ESP entries.

Method	$N = 100$	$N = 500$	$N = 1000$	f_∞ (empirical)
Full-dim GeoMed	184s	921s	>3600s	—
Full-JL (all rounds)	1.4s	7.1s	14.3s	1.00
TAVS-ESP	**0.7s**	**3.4s**	**6.1s**	**0.46**

7 Discussion

Limitations. (1) TSA excludes deployments requiring server auditability; the VRF variant in Appendix E restores verifiability at the cost of per-round proof generation. (2) The convergence proof requires honest majority $\beta < 1/2$; performance degrades in the high-β + high-ζ regime (Appendix F). (3) Interaction with client-side gradient sparsification requires separate analysis (**??**).

Scope of Paper 2. The full non-IID Byzantine convergence proof (TC2 Stages 2–4) including the non-convex extension for LLM fine-tuning and the Sleeper Agent interception experiment (E3 on LLaMA-3-8B) are developed in the companion paper: *"Convergence of Trust-Adaptive Shadow Aggregation under Non-IID Data and Byzantine Workers."* [TODO: cite companion paper when available]

8 Conclusion

We presented TAVS-ESP, the first Byzantine-robust FL framework to formally co-design verification scheduling and detection as a unified system. By grounding trust scores in JL-projected geometric distances, deriving both projection geometry and promotion assignments from a shared CSPRNG secret, and enforcing Bayesian-weighted shadow aggregation with coalition-scale budget constraints, the system achieves provably stronger security than full-verification baselines at $O(f_\infty Nk)$ steady-state cost. We believe formalizing the scheduling dimension as a first-class security primitive—with its own threat model, formal definitions, and provable guarantees—opens a new and practically important line of inquiry in Byzantine-robust federated learning.

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A Assumptions

A1 (L-smoothness): $\|\nabla F(u) - \nabla F(v)\|_2 \le L\,\|u - v\|_2$.
A2 (μ-strong convexity, Stages 1–3 only): $F(u) \ge F(v) + \langle \nabla F(v),\, u - v\rangle + \frac{\mu}{2}\,\|u - v\|_2^2$.
A3 (Bounded variance): $\mathbb{E}[\|g_i - \nabla F_i(w)\|_2^2] \le \sigma^2$.
A4 (Bounded gradient norm): $\|g_i(r)\|_2 \le G_{\max}$ (enforced by gradient clipping).
A5 (Honest majority): $\beta < 1/2$.
A6 (Gradient dissimilarity, Stage 3): $\zeta \le \bar{\zeta} < \infty$.

B Full Proof of TC1

Proof. For any fixed z with $\|z\|_2 = \delta > 0$, by the JL lemma applied blockwise: for each block m, $\mathbb{P}[\|r_m z^{(m)}\|_2^2 \ge (1-\varepsilon_m)\,\|z^{(m)}\|_2^2] \ge 1 - \exp(-\Omega(k_m \varepsilon_m^2))$. Summing across blocks and applying the union bound: $\mathbb{P}[\|R_r z\|_2^2 \ge (1-\varepsilon)\delta^2] \ge 1 - \sum_m \exp(-\Omega(k_m)) \ge 1 - M\exp(-\Omega(k/M)) = 1 - \exp(-\Omega(k))$. Since $y_r = \mathrm{ChaCha20}(K, r)$ with independent counter inputs, $R_r \perp R_{r'}$ for $r \ne r'$. The multi-round bound follows: $\mathbb{P}[\text{evade all } T] \le \prod_{r=1}^{T} \mathbb{P}[\|R_r z(r)\|_2 \le \delta/2] \le \exp(-\Omega(k \cdot T \cdot \delta_{\min}^2))$. □ □

C Full Proof of TC3

Proof. The derivations $G_1(y_r)$ and $G_2(y_r)$ are independent functions of the same pseudorandom seed. $\mathcal{A}_2$ who knows $P_r = G_2(y_r)$ but not $R_r = G_1(y_r)$: knowing P_r reveals zero information about R_r because G_1 and G_2 use independent counter offsets in the ChaCha20 stream (e.g., G_1 reads bytes 0–255, G_2 reads bytes 256–511 of the 512-bit output). Therefore, $\mathcal{A}_2$'s attack in promoted rounds is indistinguishable from a random attack with respect to evasion of the next round's projection. By TC1, any non-negligible $\delta_{\min}$ is detected with probability $1 - \exp(-\Omega(k\delta_{\min}^2))$ when verified. $\mathcal{A}_1$ who knows R_r: can craft null-space attacks for the current projection, but since P_r is unknown, cannot guarantee the crafted update falls in a promoted round. Across K promoted rounds, the probability that $\mathcal{A}_1$ and $\mathcal{A}_2$ are simultaneously successful is bounded by the product of their individual bounds. The decoy term $(1 - p_{\text{decoy}})^K$ bounds the probability that a Tier 3 promoted adversary avoids all surprise verifications over K rounds. □ □

D TC2: Full Convergence Proof

Stage 2: Trust-dynamic weights, IID. Define the composite Lyapunov potential $\Phi(r) = F(w(r)) - F^* + c\sum_i (T_i(r) - T_i^*)^2$, where $T_i^* = \mathbb{E}[\varphi_i]/(1-\alpha)$ is the steady-state trust score and $c = \eta L G_{\max}/(1-\alpha^2)$. By the trust stability lemma ($|T_i(r) - T_i^*| \le \alpha^r |T_i(0) - T_i^*| + C_\sigma$), the trust deviation term contracts geometrically. Choosing $\gamma = \min(\mu\eta, 1-\alpha^2)$ yields $\mathbb{E}[\Phi(r+1) \mid \Phi(r)] \le (1-\gamma)\Phi(r) + C_{\text{floor}}$, where $C_{\text{floor}} = \eta^2\sigma^2/N + c(1-\alpha)^2\sigma_\varphi^2$. Since $F(w(r)) - F^* \le \Phi(r)$, geometric convergence follows.

Stage 3: Non-IID + Byzantine. Define good event G_r: all Byzantine clients in $\mathcal{V}(r)$ are correctly classified as outliers. By TC1, $\mathbb{P}[G_r] \ge 1 - \exp(-\Omega(k\delta_{\min}^2))$. Choose $k = \Omega(\log(TN/\delta_0)/\delta_{\min}^2)$; by the union bound, $\mathbb{P}[\exists r : \neg G_r] \le \delta_0$. Conditional on all G_r, apply Stage 2 with additional non-IID drift term $C_\zeta = \zeta\eta|\mathcal{L}|/N$. The unconditional bound follows with high probability $1 - \delta_0$.

Stage 4: Non-convex (LLM setting). Drop A2. Convergence is to a stationary point neighborhood: $(1/T)\sum_r \mathbb{E}[\|\nabla F(w(r))\|_2^2] \le O(1/\sqrt{T}) + C'_p + C'_{\mathcal{B}}$, where $C'_p \le 2\eta L(1-\bar{p})G_{\max}^2|\mathcal{S}|/N$ and $C'_{\mathcal{B}} \le 2\eta L\gamma_{\text{budget}} Z G_{\max}^2$. Both constants are bounded by design parameters and vanish as $\gamma_{\text{budget}} \to 0$ and $\bar{p} \to 1$.

E VRF Variant for Auditable Deployments

For deployments requiring external verifiability (e.g., regulated cross-institutional FL where clients need cryptographic proof the server followed the protocol), replace ChaCha20 with a Verifiable Random Function: $(y_r, \pi_r) = \text{VRF}_{sk}(r)$. Publish π_r after gradient submission to enable client verification. All security proofs go through identically; the VRF pseudorandomness property is strictly stronger than CSPRNG unpredictability under TSA. Per-round overhead increases by one VRF evaluation and proof generation ($\approx$ 2ms on modern hardware with Ed25519-based VRF).

F Limitations and Future Work

High β + high ζ regime. When Byzantine fraction β is close to $1/2$ and data heterogeneity ζ is large, Byzantine client gradients become geometrically similar to honest non-IID clients in projected space. The block-variance normalization (Proposition 2) partially mitigates this, but detection power degrades. Future work should develop tighter detection bounds in this regime.

Gradient compression interaction. Top-K sparsification applied client-side before gradient submission may interact with the block-diagonal projection in ways not analyzed here. In particular, sparsification could

concentrate the attack signal in non-top-K coordinates; this edge case requires separate treatment.

Client churn. The trust ramp-up (Mechanism 3) creates sustained Tier 1 overhead in high-churn settings. Trust vouching (sketched in the main thesis document) is a direction for reducing this cost without compromising security.